



化学测量学—人工智能化学分析

Chapter1

AI for Analytical Chemistry

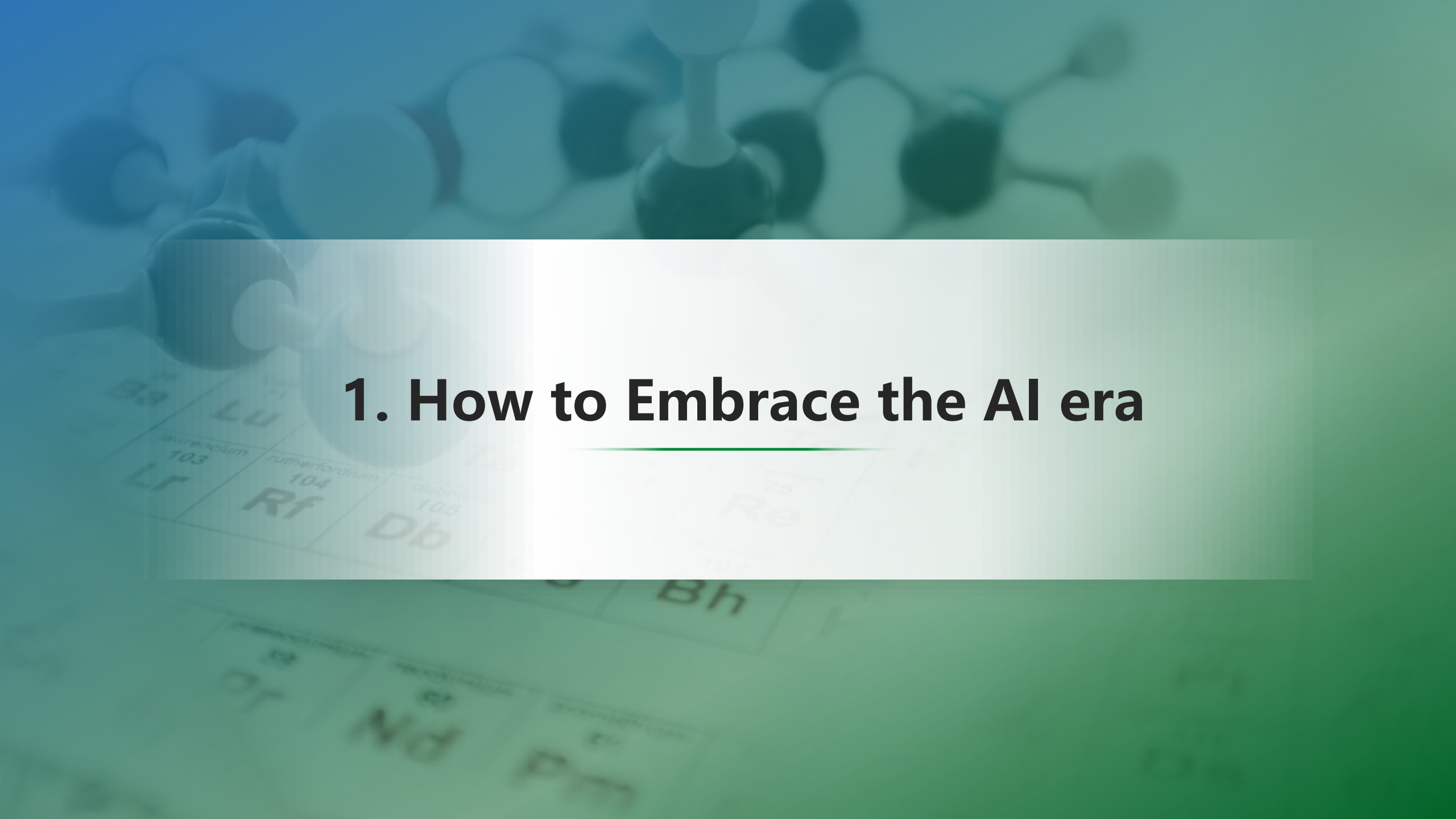
Introduction

人工智能化学分析

汪骋 (Wave) 厦门大学

Contents

- 1. How to Embrace the AI era**
- 2. Automation: Efficiency and Safety**
- 3. Data Science and Machine Learning**
- 4. About the course**



1. How to Embrace the AI era

Chemistry in the eyes of your parents



Chemistry research in reality



Modern chemical plant



？ ？ ？ ？ ？

Why is chemistry research so backward?

- Previous automation technologies did not have the flexibility to support the richness of chemistry research
- Computational technology does not have enough computing power and sufficiently accurate models to realize the "digital twin" of the chemical system

? ? ? ? ?

Five Years Later ...

Would Chemical Research Be the Same?

AI-enabled automation technologies are becoming more and more flexible; computing models are becoming more and more accurate; computing is getting faster and faster; and all technologies are developing rapidly and iteratively

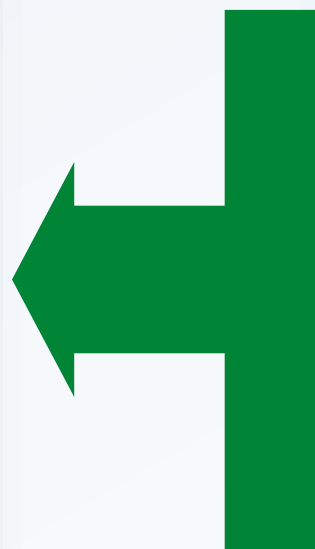
AI Reshapes Chemical Research

Modern Chemistry

合成 Make

测量 Measure

模拟 Model



AI

Multi-modal large language models LLM

Automation AutoGPT

Artificial general intelligence AGI

- Information integration
- Rule Extraction
- Generative model
- AI decision-making
- Computation acceleration

Statistical Methods in Analytical Chemistry

Chemometrics

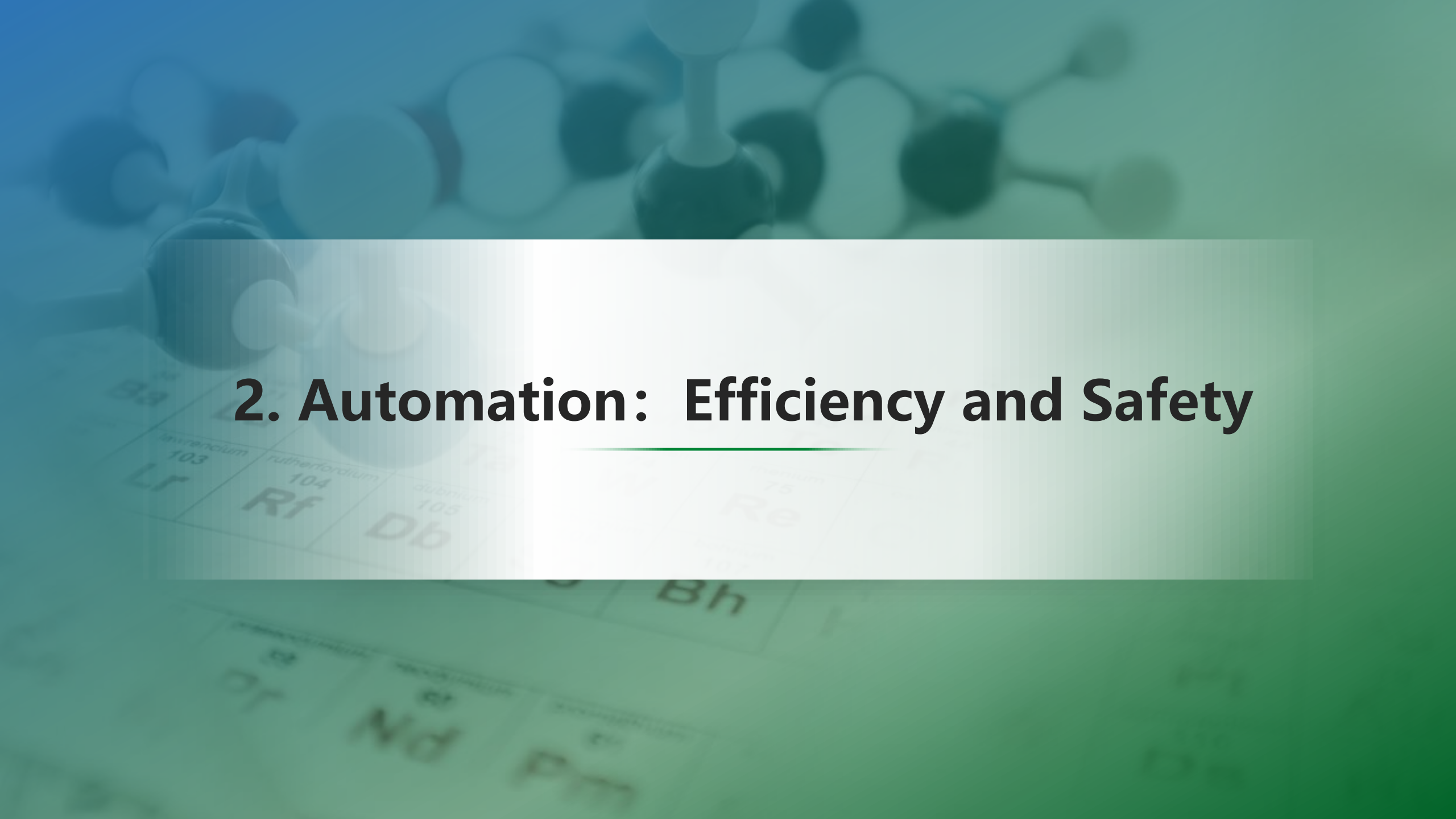
Cheminformatics

What is really new about AI4AC ?

- **More flexibility in automation and data analysis**
- **Data-driven feature extraction in neural networks**
- **Large Models as new forms of knowledge graph (the model has memory and deduction power)**

What can we expect ?

- **Automated Labs**
- **Integrating Information from multiple analytical methods (multi-modal)**
- **Knowledge-enhanced data analysis**
- **Large model as co-pilot**

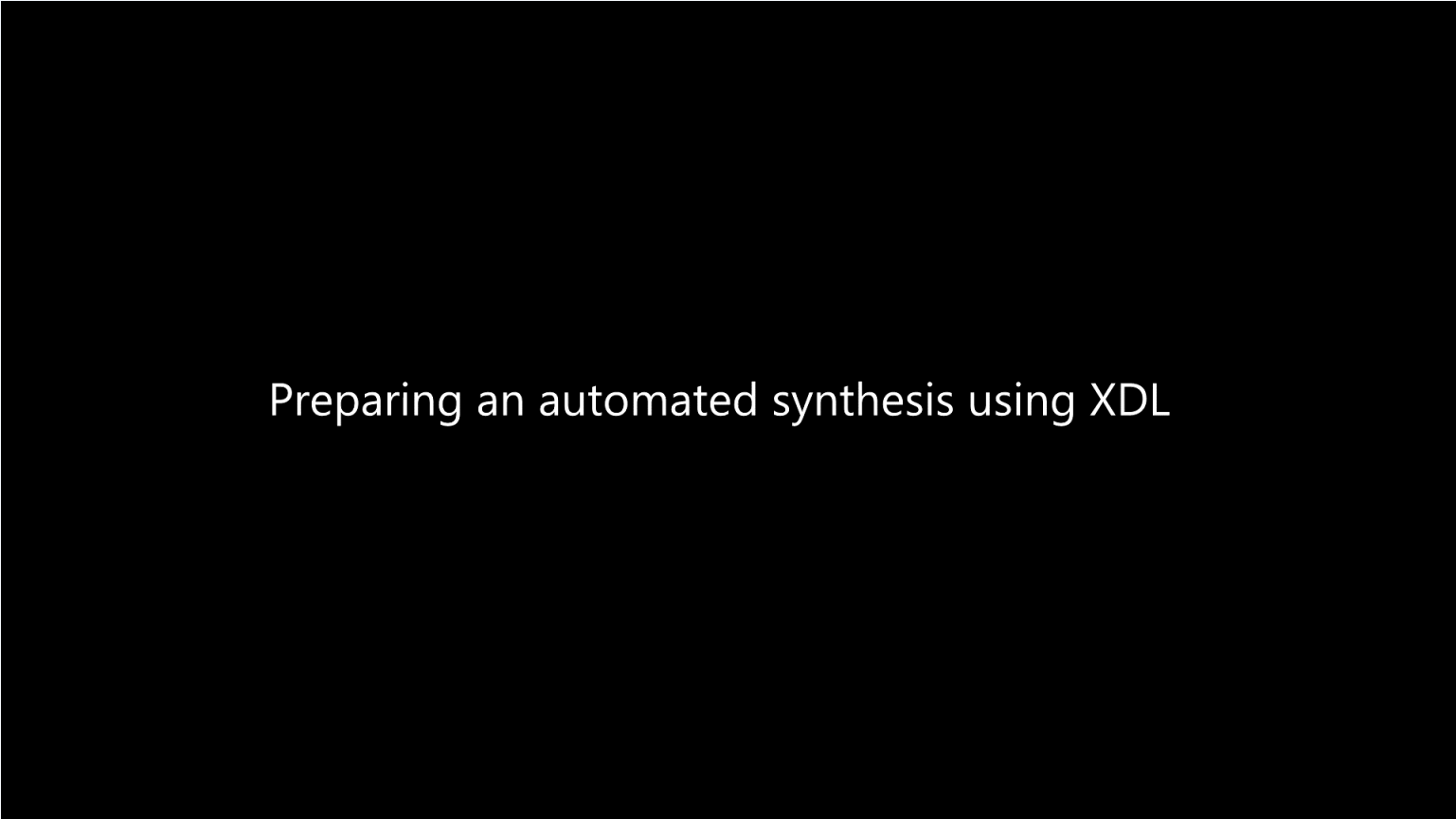


2. Automation: Efficiency and Safety

A few years ago, when "artificial intelligence" was still only on PPT ...

Adjust the experiment with code

Automation turns chemistry experiments into programming



Preparing an automated synthesis using XDL

With anthropomorphic robots

Increase flexibility and versatility and reduce the cost of developing automation equipment



[*Nature*](#) volume 583, pages237–241 (2020)

Work day and night, all year round



[*Nature*](#) volume 583, pages237–241 (2020)

Guess what is this process?



- It's anthropomorphic, but it doesn't have to be stuck to traditional chemical solutions. Vibration is used instead of magneton stirring to simplify operation
- For process optimization, the best automated operation is to avoid it

[*Nature*](#) volume 583, pages237–241 (2020)

Automation = High Throughput ?

The purpose of automation?

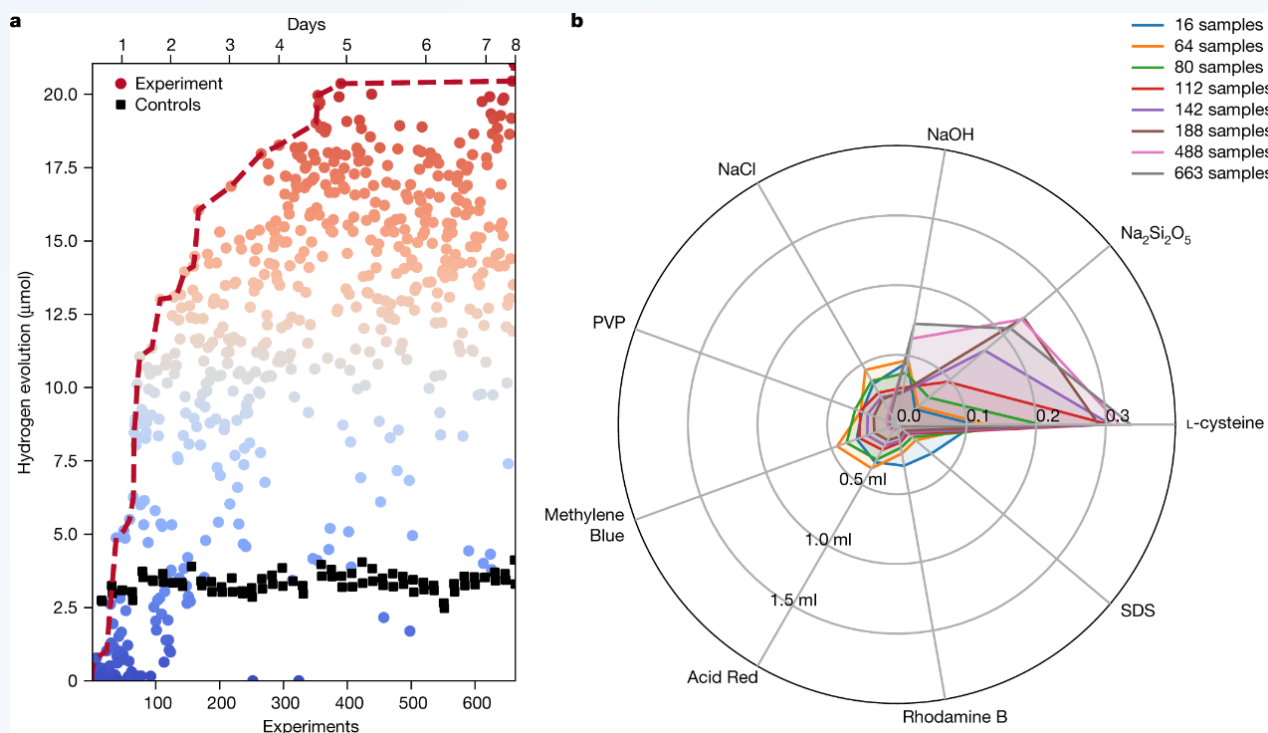
- Faster, more accurate, and safer
- Repeatable, Standard data production
- Save Manpower

Principles of chemical automation process design

- Be clear about your purpose
- Find the rate-limiting step and consider it in the design
- The shorter the process, the more reliable the automation
- Feedback iteration is required
- Consider the cost

Chemical automation combined with Bayesian optimization

Optimization of photocatalytic conditions



[Nature](#) volume 583, pages237–241 (2020)

Bayesian optimization is a method for optimizing complex functions, especially for the optimization of chemical reaction conditions.

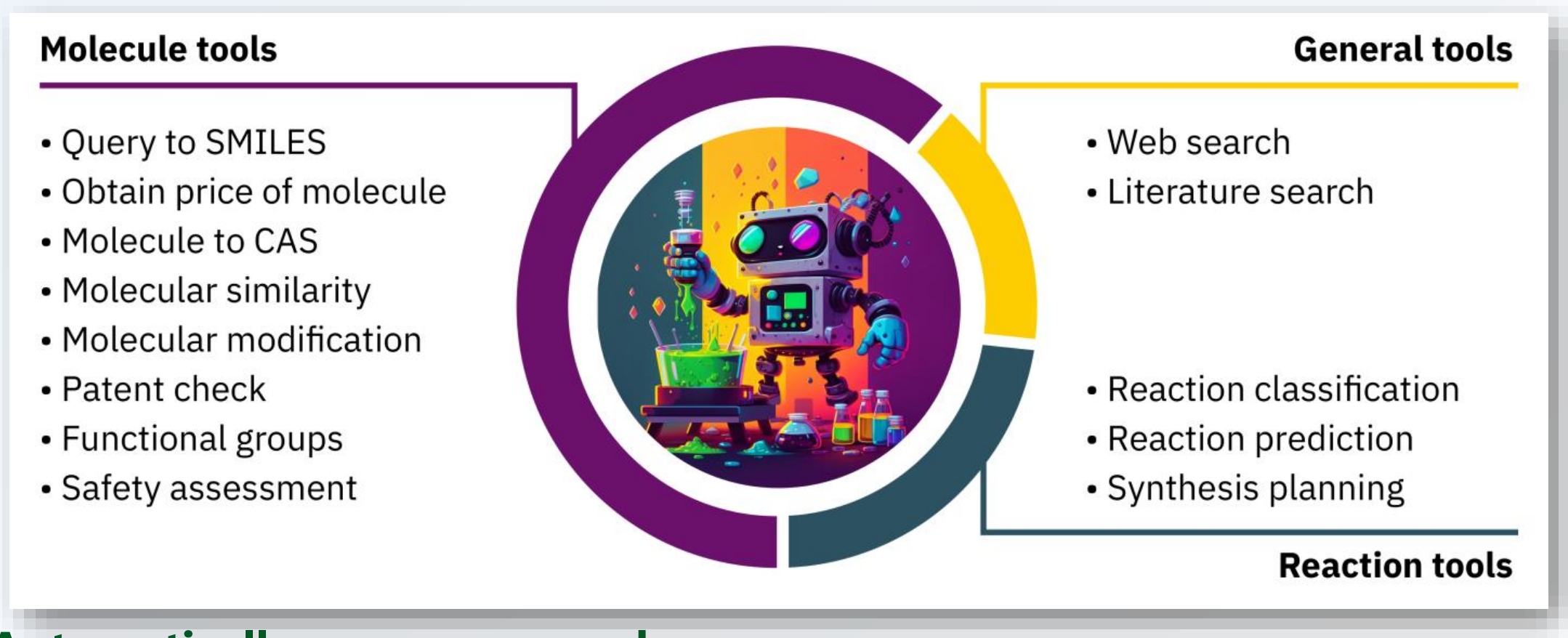
Bayes' theorem: A mathematical formula that describes the probability of an event occurring under known conditions, updating beliefs about a hypothesis based on prior knowledge and new evidence.

Combination of Bayesian optimization and automatic iteration: firstly, a small number of initial experimental points are obtained (random, uniform, Latin hypercube sampling, etc.), and then the model is established, the performance and uncertainty of the remaining conditions are predicted by Bayes' theorem, sampling is carried out according to the optimization goal to continual with the iteration.

**When "artificial general intelligence" is already
on the horizon ...**

A chemical synthesis system driven by large language models

Amazing iteration speed



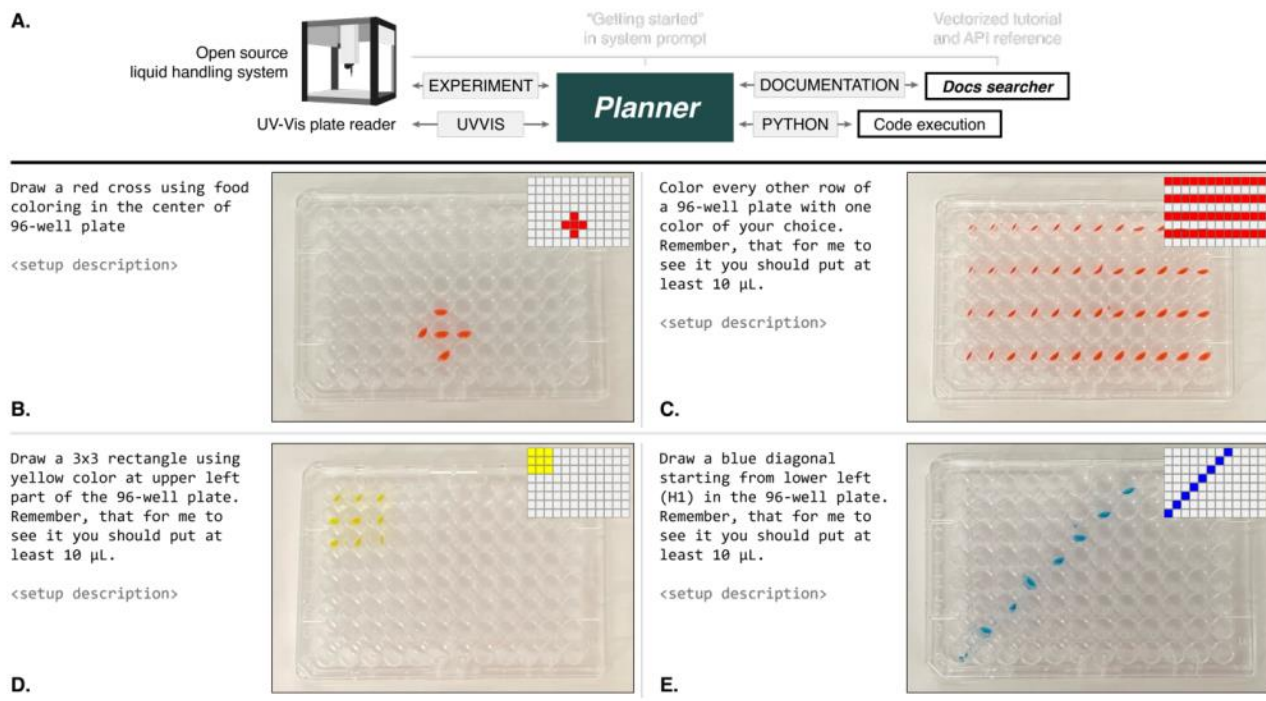
Automatically search chemistry knowledge from the web and literature, complete tasks, and evaluate them

<https://doi.org/10.48550/arXiv.2304.05376> Tue, 11 Apr 2023

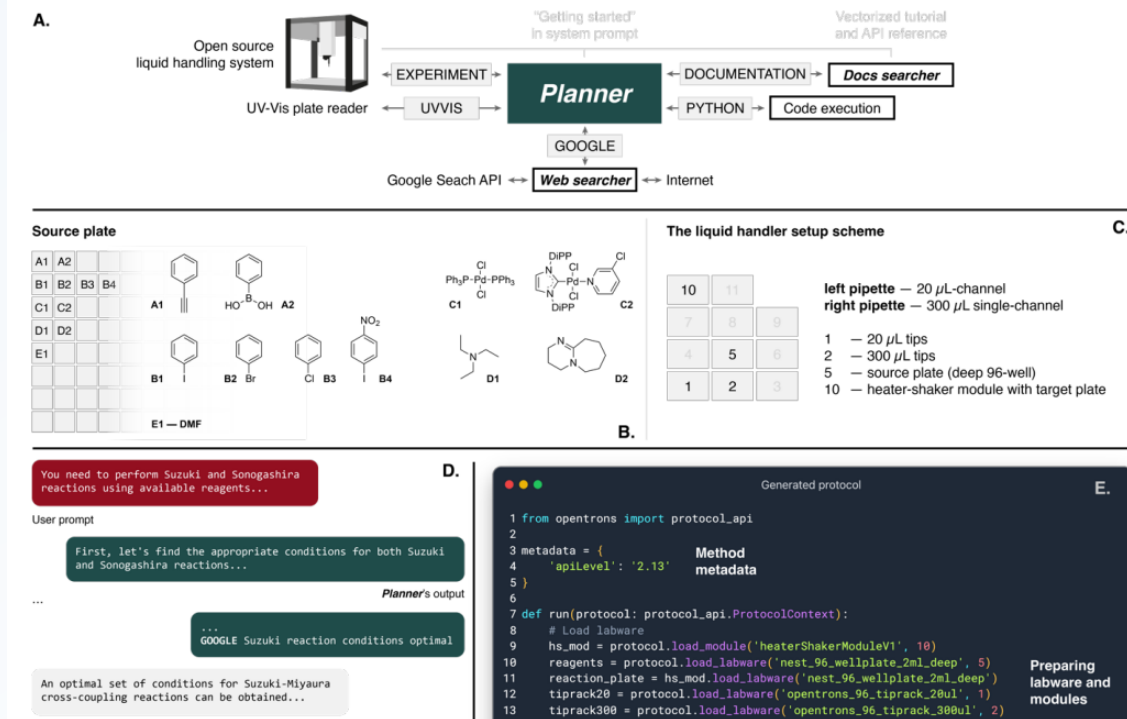
Open AI GPT-4 发布 , March 14, 2023

A chemical synthesis system driven by large language models

Start by reading the instructions to control the hardware device



Search, plan, and test reactions autonomously



Experiment with intelligent decision-making and device calling

<https://doi.org/10.48550/arXiv.2304.05332> Tue, 11 Apr 2023

Open AI GPT-4 发布, March 14, 2023

Large Language Models

In the field of general artificial intelligence, large language models are an important step towards more advanced, adaptive, and understanding AI systems. These models, by learning and processing a vast amount of textual data, can not only understand and generate complex natural language but also exhibit flexibility and creativity across a variety of tasks and scenarios. They are capable of context analysis, sentiment assessment, and logical reasoning, demonstrating language processing capabilities that are close to or even surpass human levels in some aspects.

Large language models demonstrate enormous potential in scientific research and technological development. By analyzing and understanding a vast amount of scientific literature, research reports, and professional articles, these models can assist scientists in discovering new research directions, accelerating the data analysis and theoretical deduction process. In fields such as chemistry, physics, biology, and others, large language models can be used to predict experimental results, propose new research hypotheses, and even assist in designing experiments and interpreting experimental data.



Task:

I need to synthesize a sample of atorvastatin. Please tell me how to synthesize it. Then tell me how much will it cost to buy all the reactants I need. Provide specific substance names (iupac) or smiles. If you can't determine the price of the reactants, try another synthetic route.

Both models demonstrate molecular synthesis planning capabilities

ChemCrow:

First, I need to find the SMILES of atorvastatin to plan a synthetic route.

Action: Get SMILES of molecule

Action Input: atorvastatin

Observation:

CC(C)c1c(C(=O)Nc2ccccc2)c(-c2ccccc2)c(-c2ccc(F)cc2)n1CC[C@@H](O)C[C@@H](O)CC(=O)O

Thought: Now that I have the SMILES of atorvastatin, I can plan a synthetic route.

Action: Molecular synthesis planner.

Action Input:

CC(C)c1c(C(=O)Nc2ccccc2)c(-c2ccccc2)c(-c2ccc(F)cc2)n1CC[C@@H](O)C[C@@H](O)CC(=O)O

Observation: To make atorvastatin with a molecular weight of 558.65, density of 997.77, and a product mass of 500 mg, follow these steps:

1. Add 0.7 mmol of (3R,5R)-7-[2-(4-fluorophenyl)-5-isopropyl-3-phenyl-4-phenylcarbamoyl-pyrrol-1-yl]-3,5-dihydroxy-heptanoic acid, t-butyl ester to the reaction vessel.
2. Add 0.1 ml of dichloromethane to the reaction mixture.
3. Add 0.2 mmol of trifluoroacetic acid (TFA) to the reaction mixture.
4. Stir the reaction mixture at normal stirring speed and at a temperature of 25°C for 1 hour (3600 seconds).
5. Concentrate the reaction mixture.
6. Purify the product (Note: The action type Purify cannot currently be converted to a Chemspeed instruction).
7. Store the final product with the sample name:

CC(C)c1c(C(=O)Nc2ccccc2)c(-c2ccccc2)c(-c2ccc(F)cc2)n1CC[C@@H](O)C[C@@H](O)CC(=O)O

Thought: I have a synthetic route for atorvastatin, now I need to find the SMILES of the reactants and their prices.

Action: Get SMILES of molecule

Action Input:

(3R,5R)-7-[2-(4-fluorophenyl)-5-isopropyl-3-phenyl-4-phenylcarbamoyl-pyrrol-1-yl]-3,5-dihydroxy-heptanoic acid, t-butyl ester

Observation:

CC(C)c1c(C(=O)Nc2ccccc2)c(-c2ccccc2)c(-c2ccc(F)cc2)n1CC[C@@H](O)C[C@@H](O)CC(=O)OC(C)(C)C

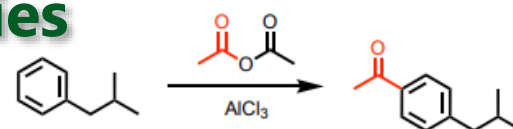
Thought: Now I have the SMILES of the first reactant, I will find its price.

Action: Get price of molecule

Action Input:

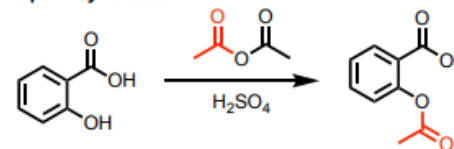
CC(C)c1c(C(=O)Nc2ccccc2)c(-c2ccccc2)c(-c2ccc(F)cc2)n1CC[C@@H](O)C[C@@H](O)CC(=O)OC(C)(C)C
Obtaining data for 1 substances.

A. Ibuprofen synthesis

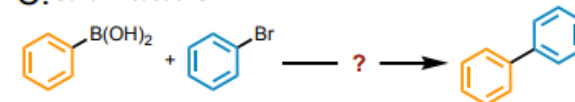


Agent correctly identified the first step (Friedel-Crafts acylation) in the synthesis of ibuprofen.

B. Aspirin synthesis

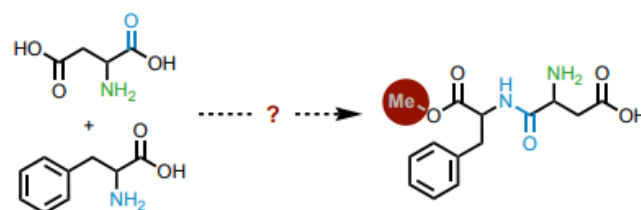


C. Suzuki reactions



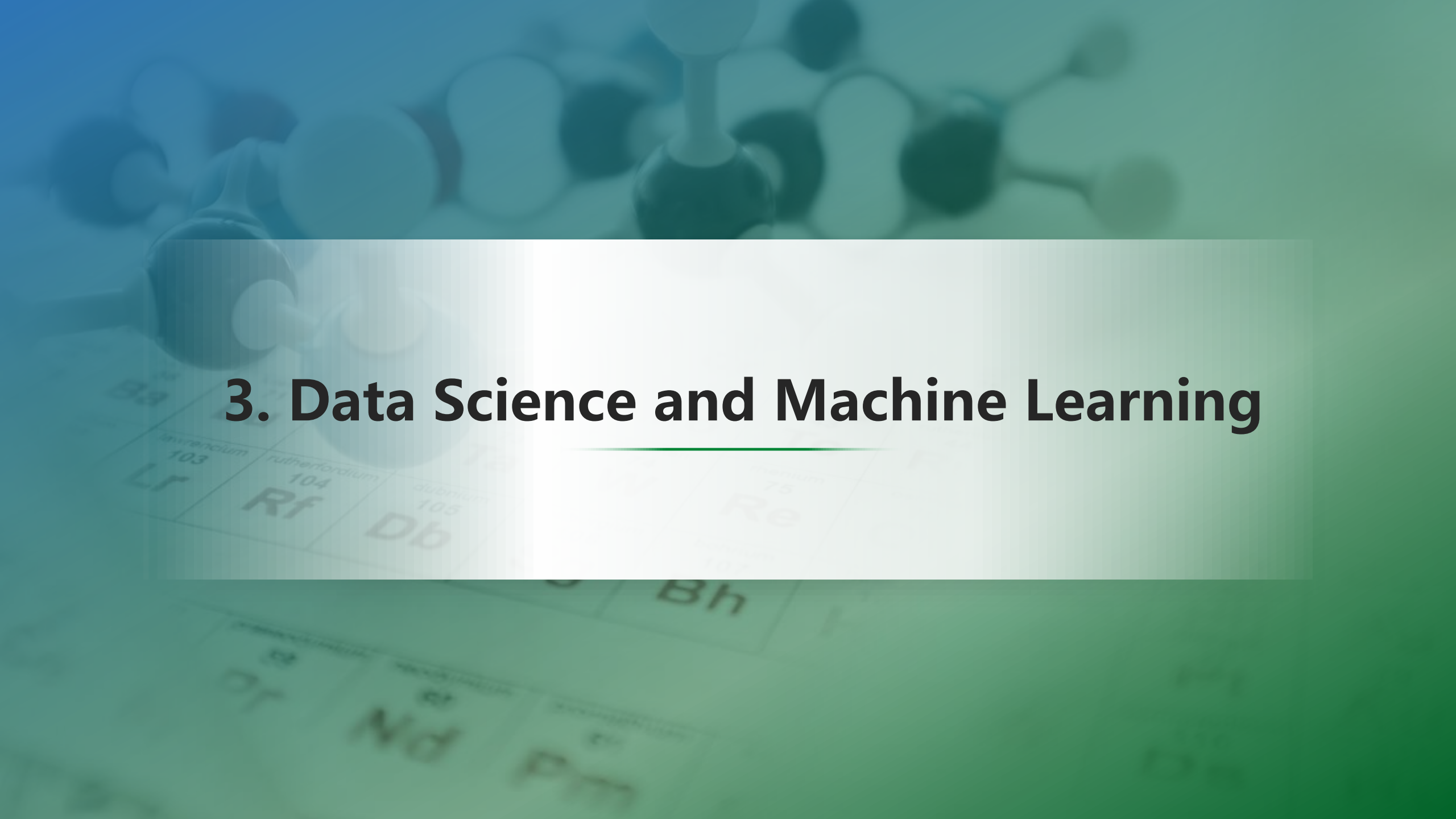
No reaction conditions, but finds information about them. Observed correct choice of catalyst and base.

D. Aspartame synthesis



No reaction conditions and missing source of "methyl" group necessary to make aspartame.

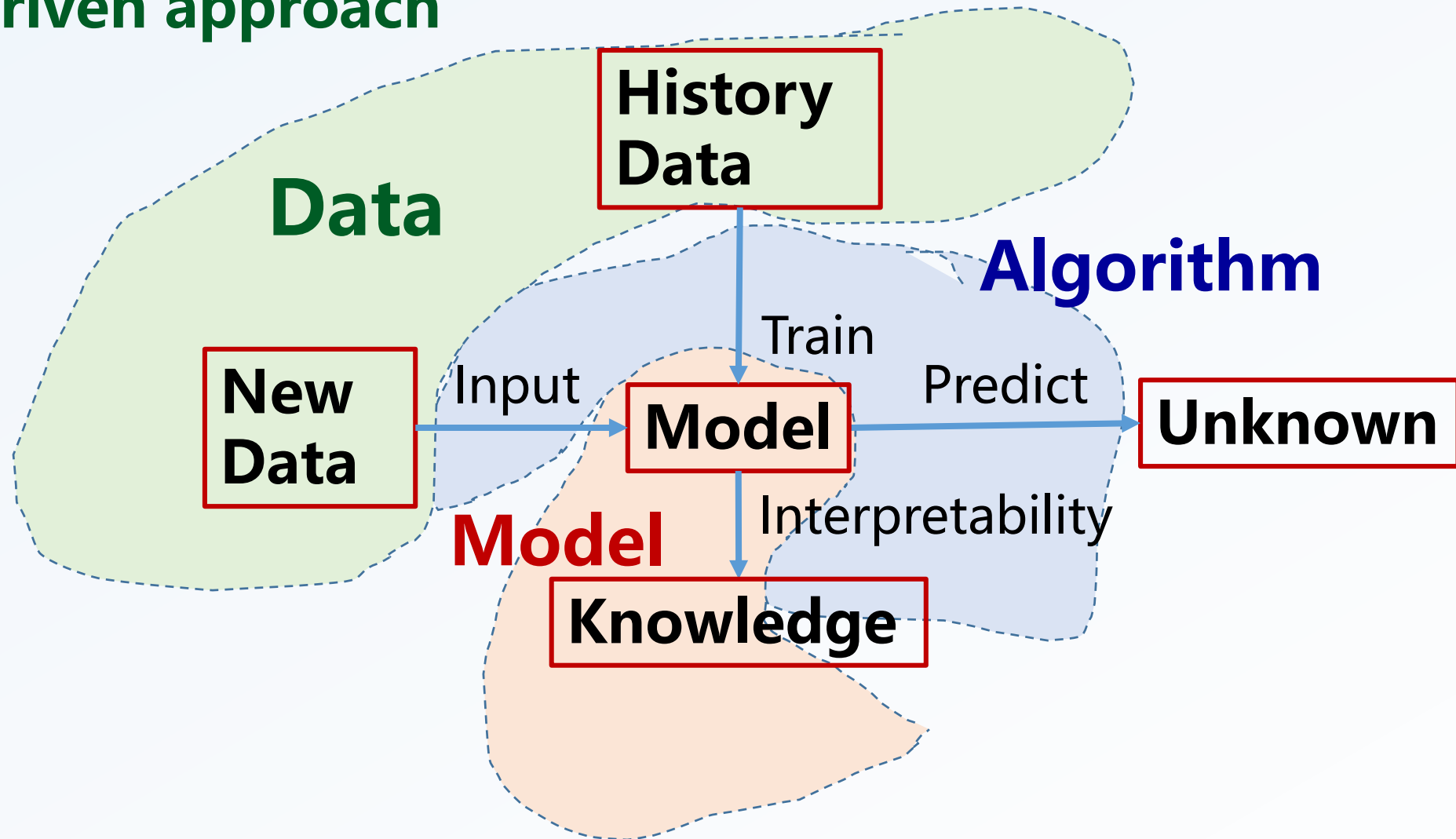
Figure 2. Agent's capabilities in the synthesis planning task. A. Ibuprofen synthesis. B. Aspirin synthesis. C. Suzuki reaction mechanism study, where the Agent had to choose how to study the mechanism. D. Aspartame synthesis.



3. Data Science and Machine Learning

What is machine learning?

A data-driven approach



Different from other approaches?

Hypothesis-driven approach

confirmed or falsified



Failures

- {

No hypothesis can be made

It takes a lot of computing power to come up with a hypothesis
- {

Variables are numerous and inseparable

Intermediate process measurement is difficult

The cost of the experiment is too high

Key elements for Data Sciene

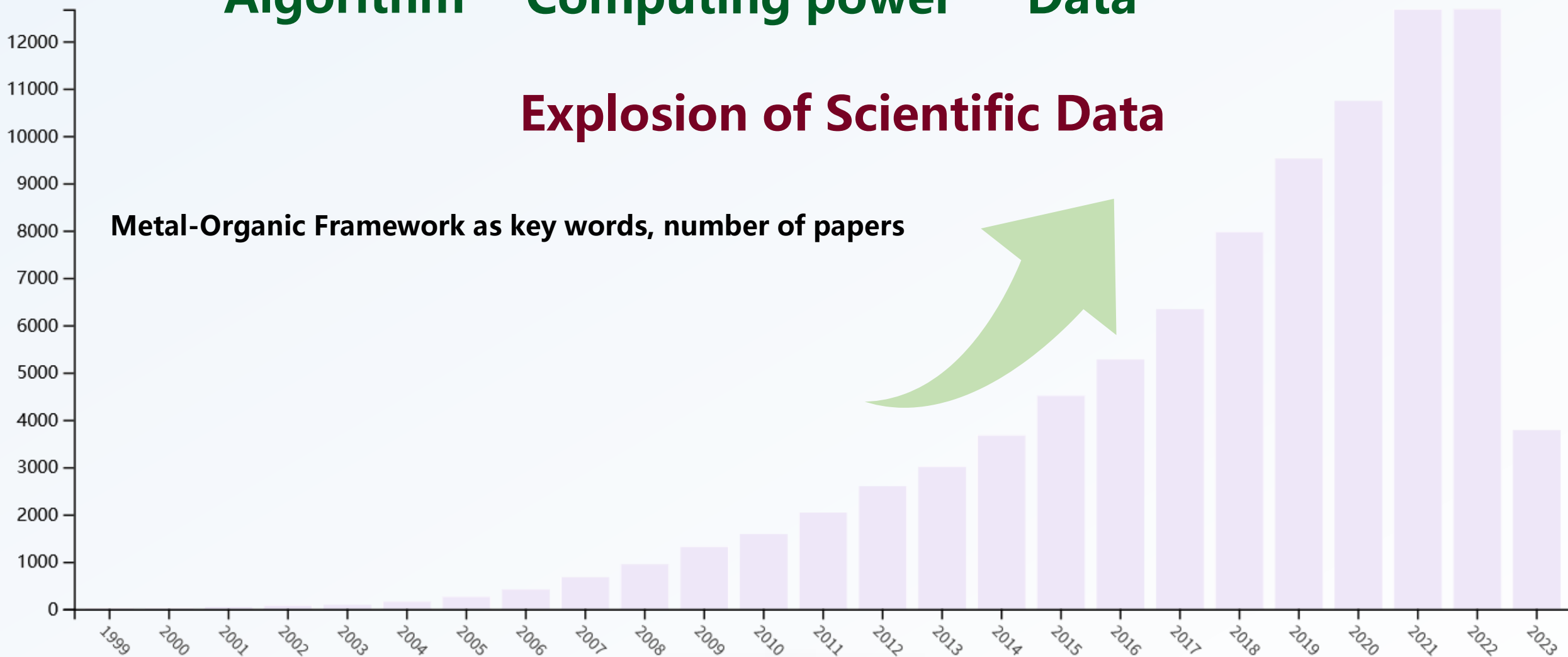
Algorithm

Computing power

Data

Explosion of Scientific Data

Metal-Organic Framework as key words, number of papers



Data Collection

■ Automation

■ Data Extraction from Literature

- Large language models and other NLP methods have made significant breakthroughs in text information extraction
- The multi-modal model has been born, and the extraction of scientific image information is a difficult hurdle to be overcome
- Chemical-specific information extraction such as molecular structure recognition is about to be overcome

■ Database

- Multiple commercial and open-source databases have accumulated a large amount of standardized experimental and computational data
- Data sharing and integration still face many challenges beyond science

Algorithm

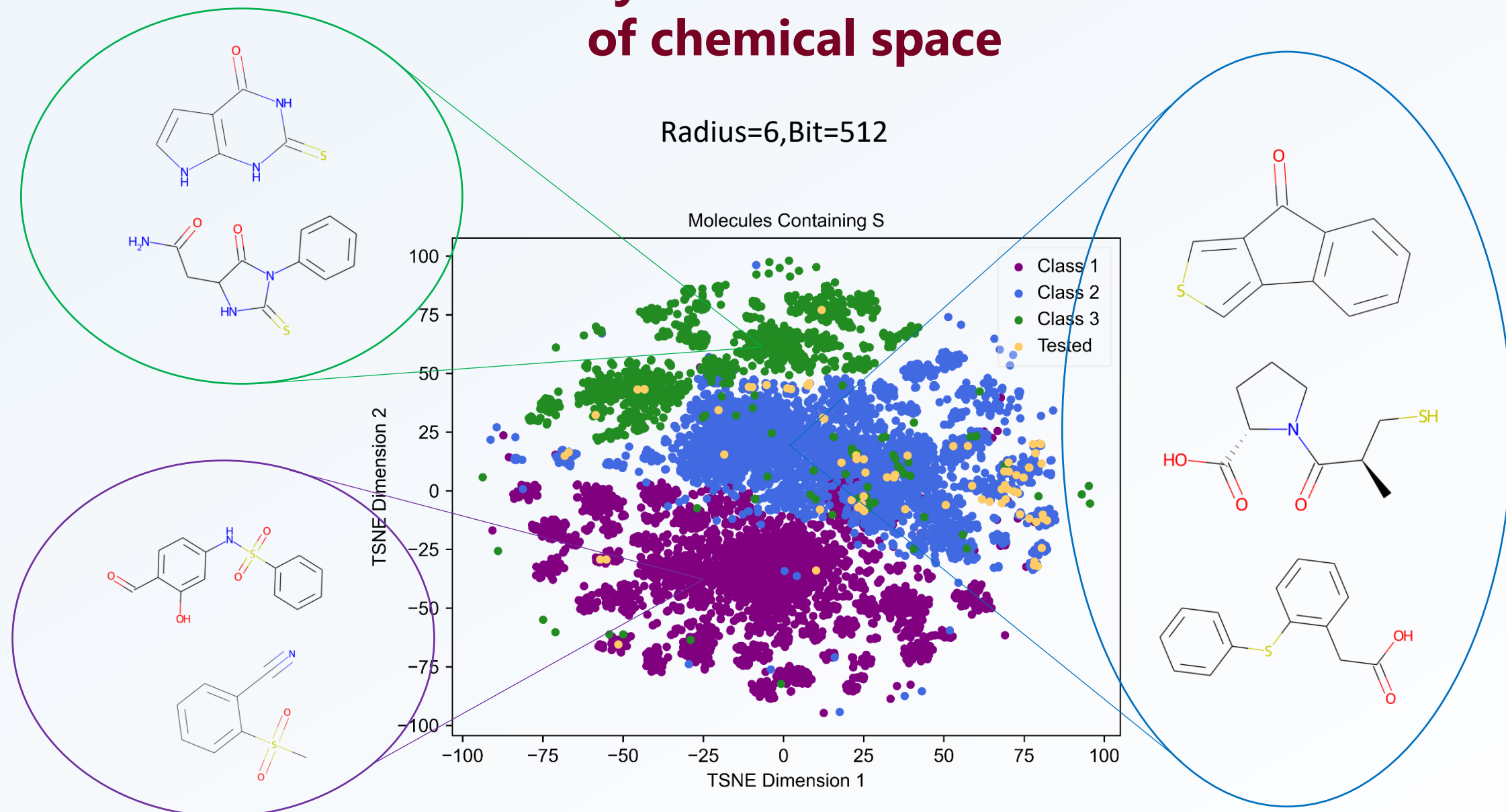
Feature Extraction

How to describe molecules? (a few examples)

- **SMILES**: String description, similar to system naming
- **Morgan Fingerprint** : describe the molecule with the presence or absence of fragments
- **Morfeus**: Molecular structure information and properties are calculated and extracted as descriptors
- **Unimol**: Pre-trained molecular models that describe molecules with vectors
- **Graph model**: Molecules as mathematical graphs (atoms as nodes, bonds as edges)

Systematic search of chemical space

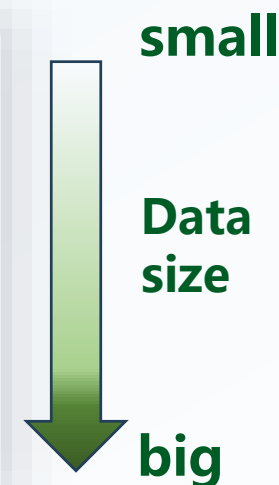
Radius=6, Bit=512



Algorithm

Model for big and small datasets

- Random Forest, XGBOOST, LightBGM
- Gaussian process model
- Graph convolutional neural networks
- Molecular pre-trained models and transfer learning

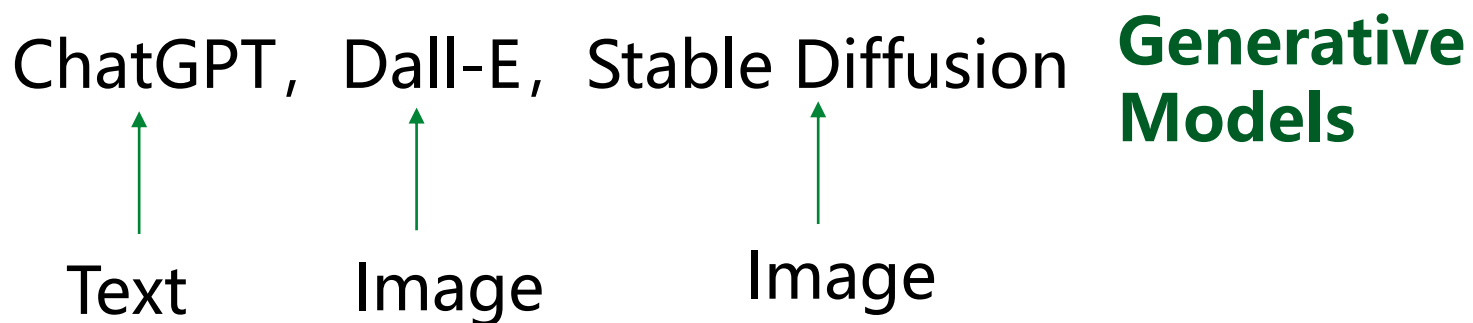


Interpretability

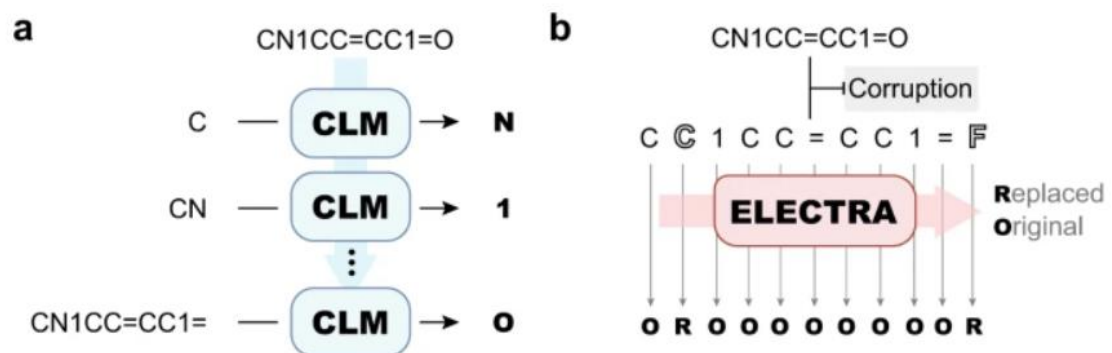
- Feature importance
- Shapley value: from game theory, analyzing the contribution of each feature to the outcome
- Attention: Neural network techniques that make the model more focused on the important parts of the input data

Algorithm

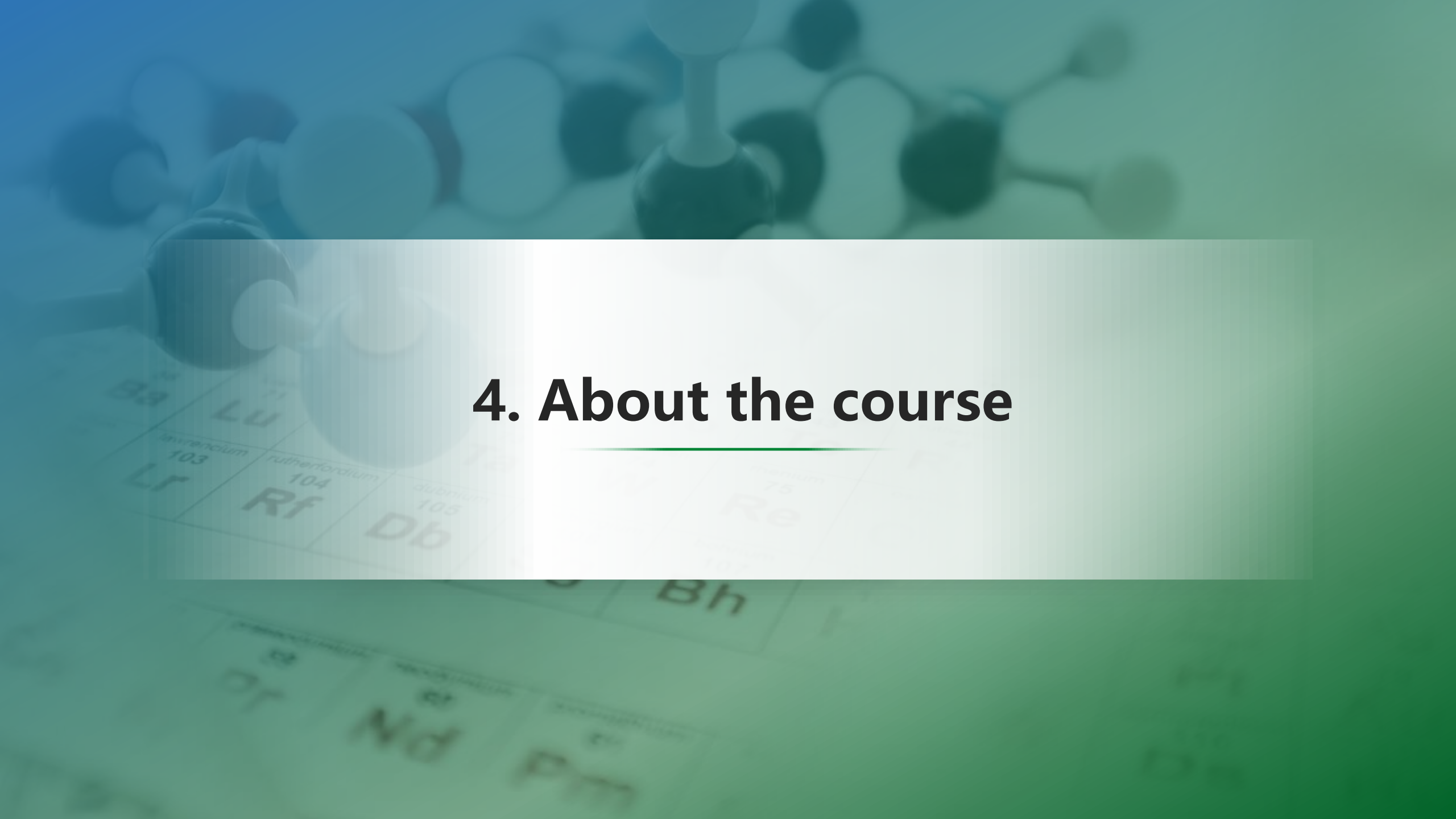
Molecular generation



Medicinal molecule generation



Nat. Commun. 2023, 14, 114



4. About the course

Contents

- **Basic Machine Learning and Chemometrics**
- **AI Application in Analytical Chemistry**
- **Homework as practical**
- **Term Project**

**Term Project must include data collection
(possibly from literature)**

Term Project can lead to a publication

Possible Topics for Term Paper

- **Vibrational frequency prediction (limited to a certain category of compound, e.g. NCS^- as a ligand)**
- **NMR prediction (limited to a certain category of compound, e.g. chemical shift change after coordination)**
- **EI-MS prediction (data available)**

Requirements

- **Attend Class**
- **Finish Homework**
- **Focus on Term Project**
- **Learn from field**

Homework 1

- **Install anaconda, jupyter notebook**
- **Write a “hello world” code**

Future
is now

